

Sequential Logic Circuits

Registers

• Group of many flip/flops to store multiple bits of information is called as "Register".

• Can used to perform diff func

1) Data storing

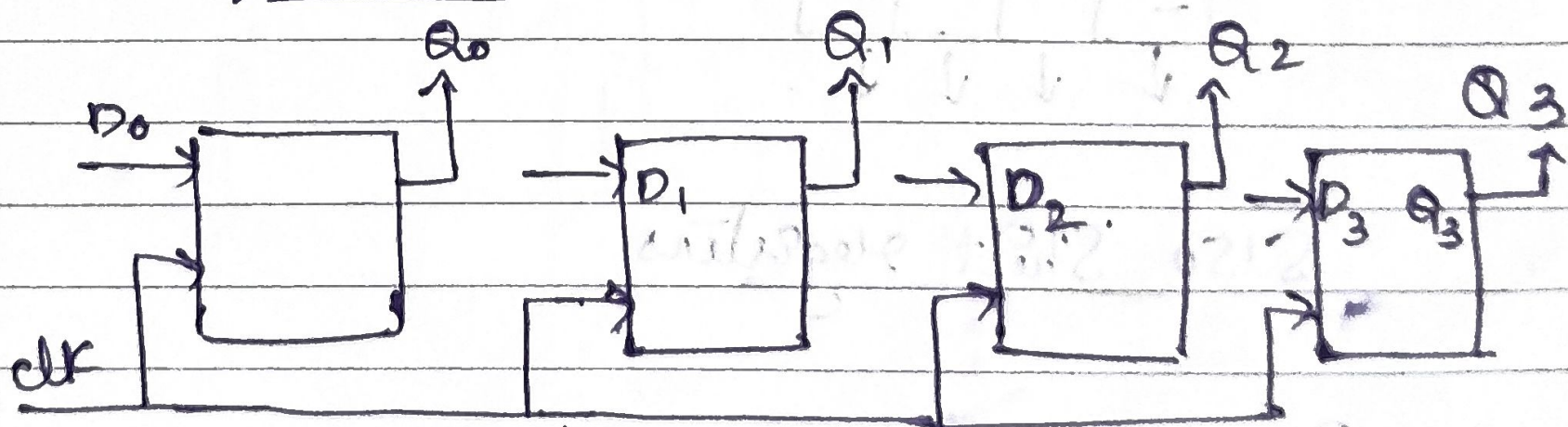
2) Data shifting

Registers are classified as

1) Buffer Registers

2) Shift Registers

1) Buffer Register



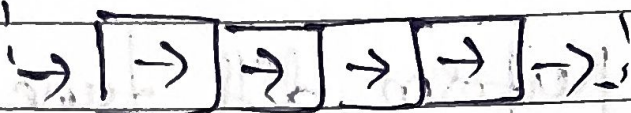
holds 4-bit data

Serial shift registers

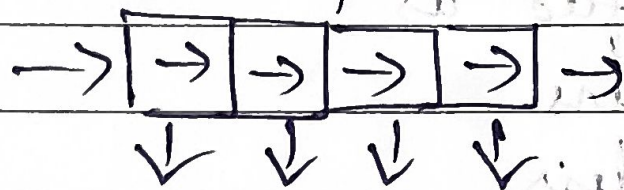
Serial in serial out (Right shift)



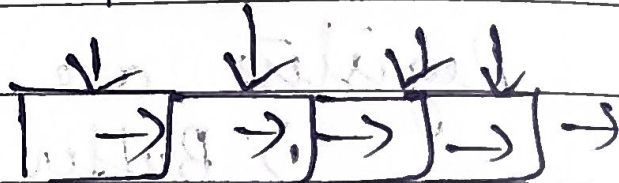
Rotate right

Parallel shift registers

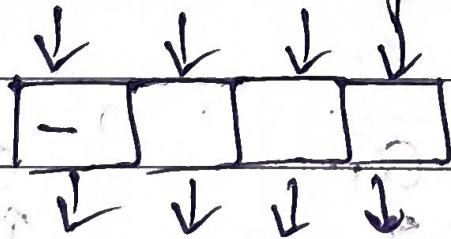
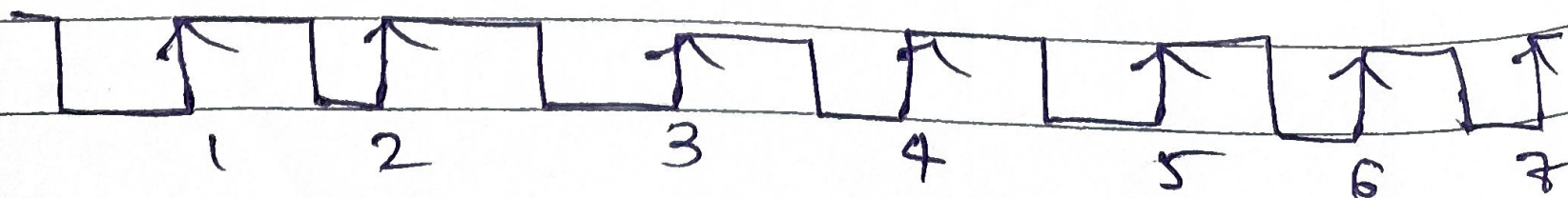
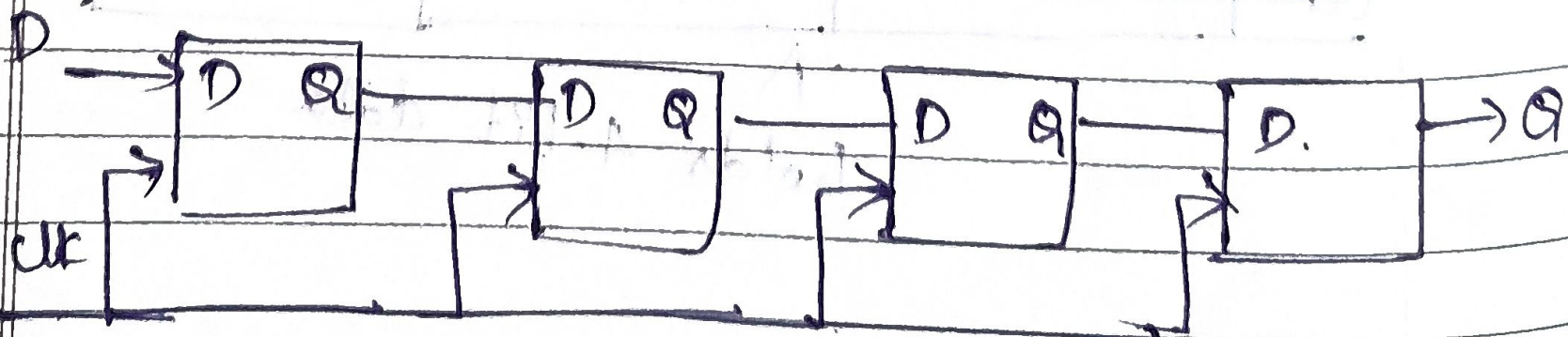
Serial in parallel out



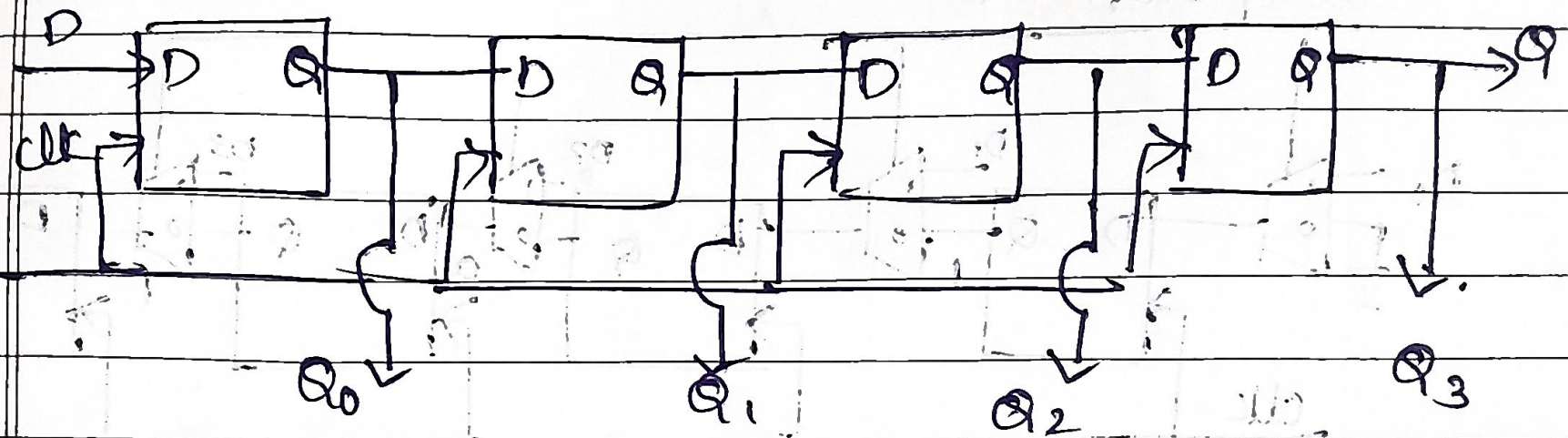
Parallel in serial out



Parallel in parallel out

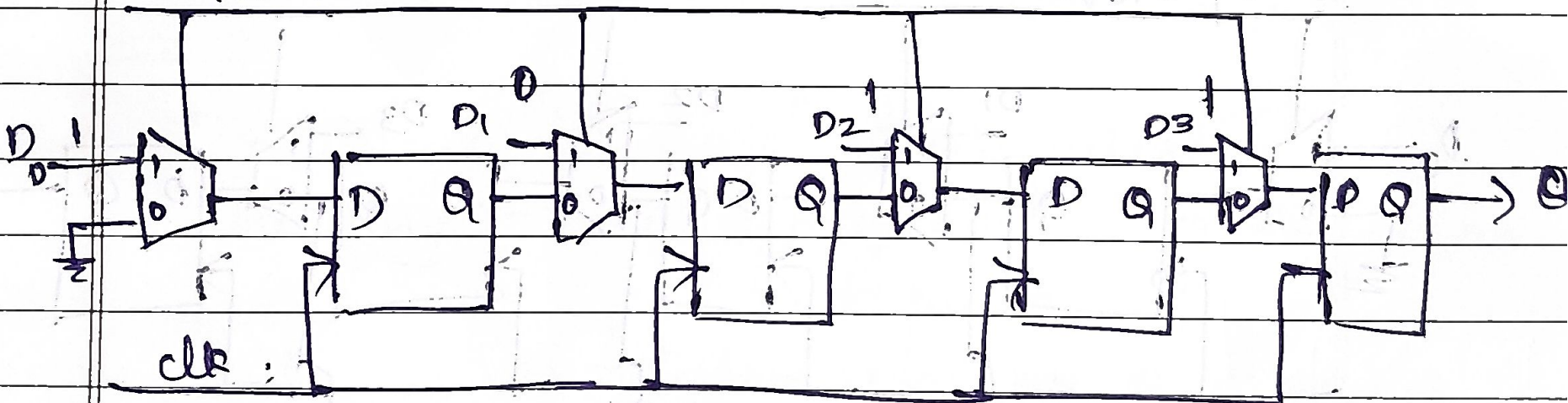
SISO Shift registers

SISO Shift registers



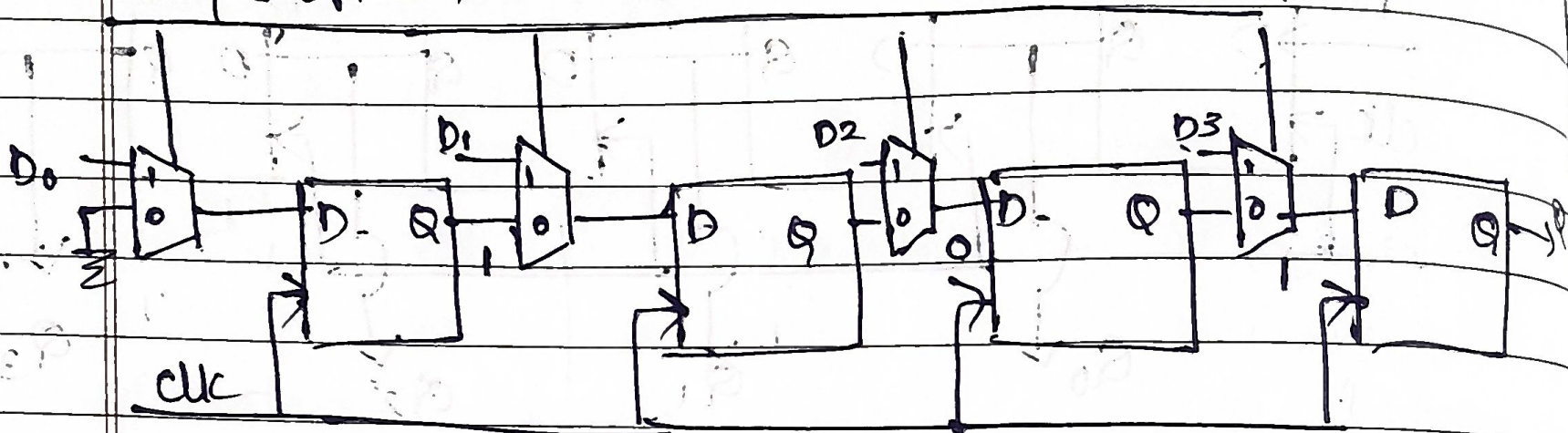
PISO shift registers

Load/Shift = 1



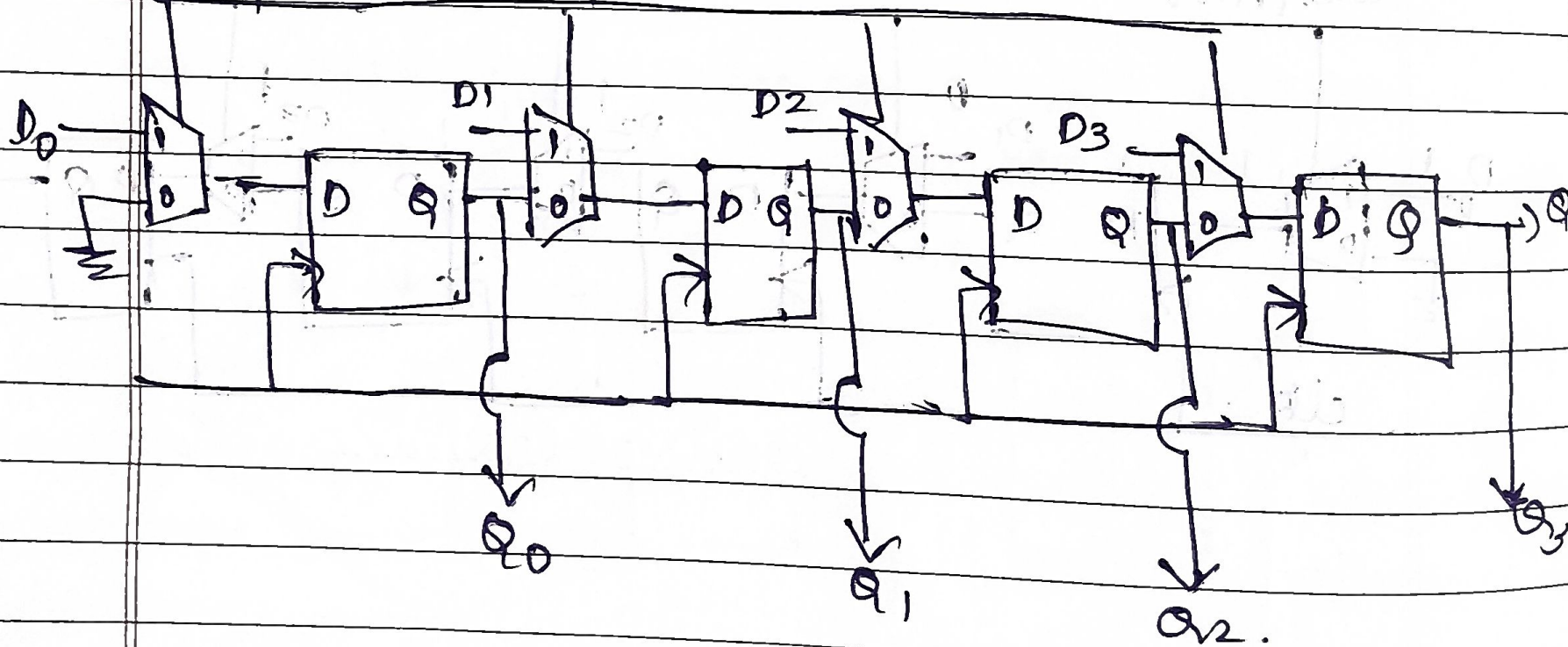
PLSO Shift Register

load / $\overline{\text{shift}} = 0$

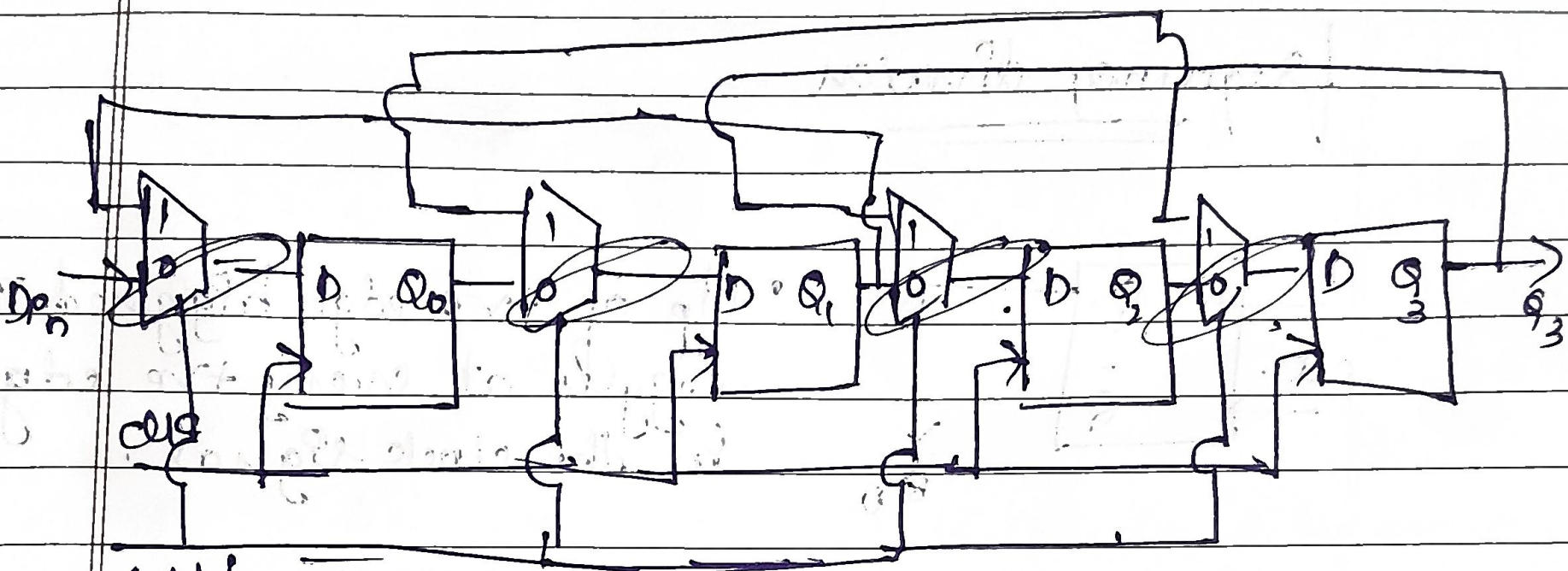


PIPO Shift Register

load / $\overline{\text{shift}}$



Bidirectional Shift registers



left/right = 0.

Counters

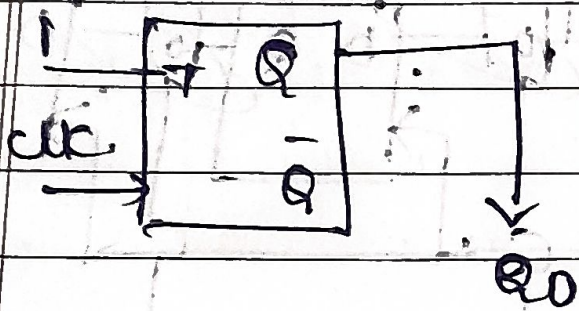
- It counts the no of clock pulses.
- Counting → Sequential manner to count no of clock pulse
- Random manner to generate a specific sequence.

Classification.

- Clock - Synchronous or Asynchronous
- clock trigger - posedge / negedge
- counts - binary, Decode
- Count direction - up, down, up-down, Random

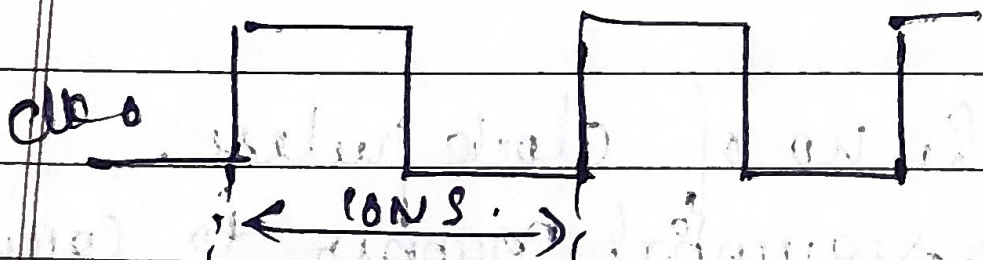
- flipflops - JK or T or D.

frequency division

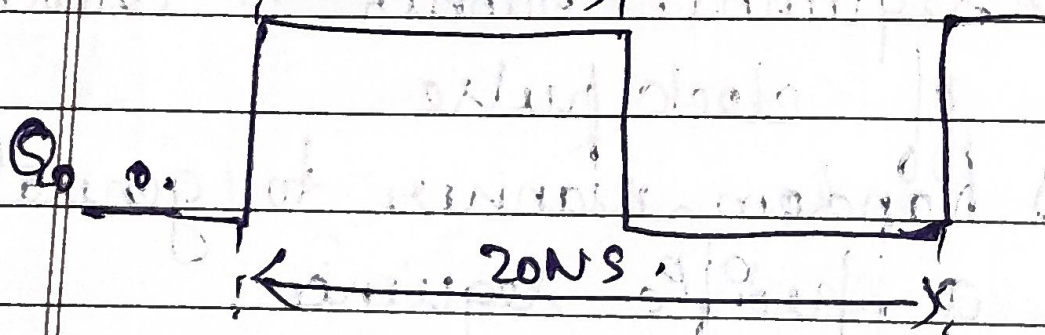


- o/p of the edge triggered TFF toggles at every the edge in the clock signal.

- Remains constant during the period other than the instant of the edge of clock.



$$f_{clk} = \frac{1}{10\text{ns}} = 100\text{MHz}$$



$$f_{Q0} = \frac{1}{20\text{ns}} = 50\text{MHz}$$

$$f_{Q0} = \frac{f_{clk}}{2}$$

Synchronous up Counter

up Counting sequence

$Q_2 \quad Q_1 \quad Q_0$

0 0 0

0 0 1

0 1 0

0 1 1

1 0 0

1 0 1

1 1 0

1 1 1

• Q_0 toggles for every cycle

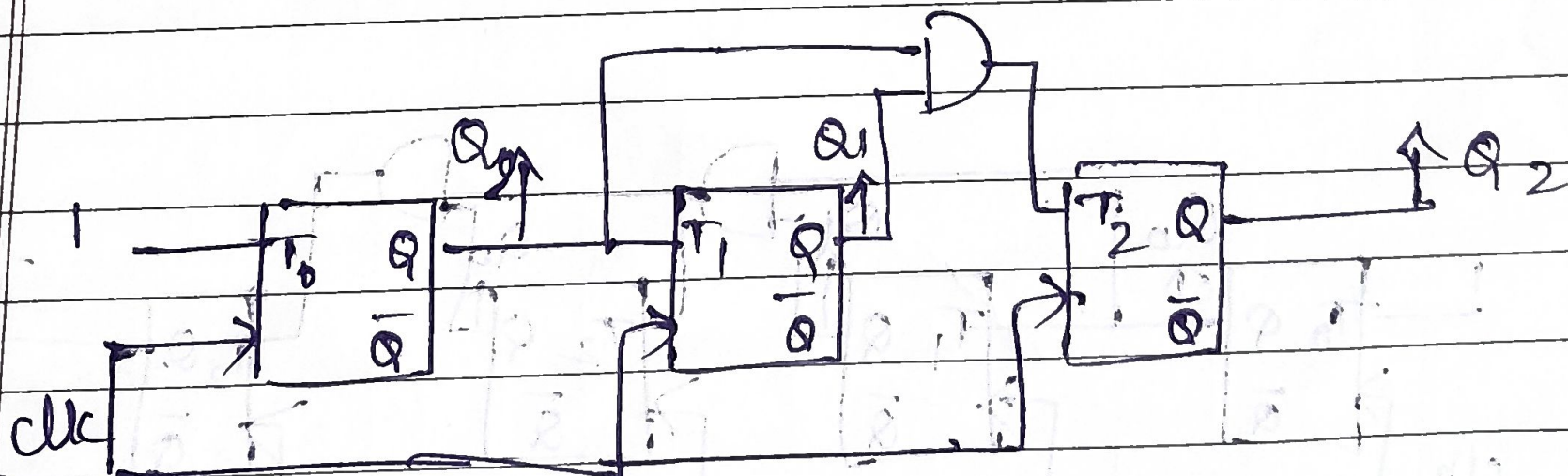
• Q_1 toggles only when Q_0

Q_2 toggles only if $(Q_0 \neq Q_1)$

are 1

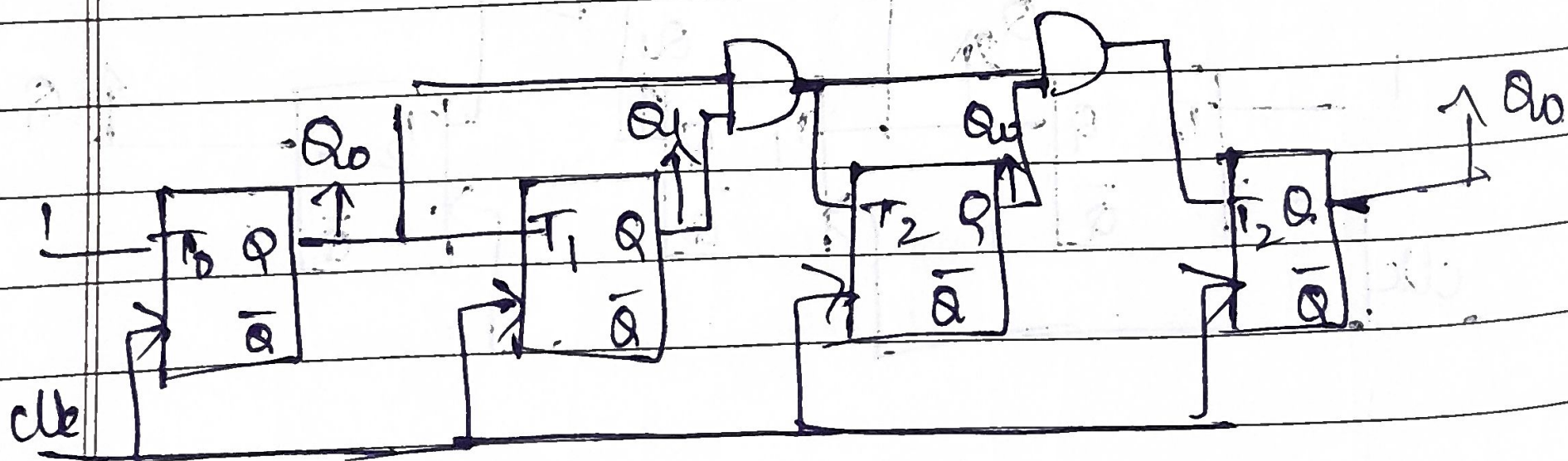
• T FF toggles only when

the T input is 1

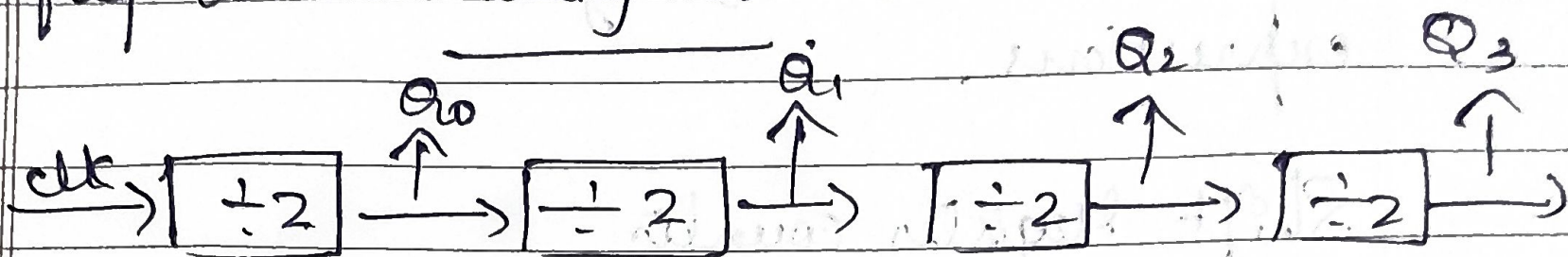


Modulus of a Counter

- The number of unique states that a counter may have before the sequence repeats itself is the modulus of the counter.
- An up counter with 3 FF counts from 0 to 7 and again repeats from 0 to 8. So, the no of states is 8.
- So, a counter with 'n' no of FF counts from 0 to $2^n - 1$. So, the modulus of this counter is 2^n .
- The o/p frequency of the last FF of any counter will be the clock frequency divided by the mod of the counter.
- A T FF with T i/p tied to VCC acts as divide by 2 counter.



freq division using counter



$$f_{Q_0} = \frac{1}{2} f_{clk} \quad f_{Q_1} = \frac{1}{4} f_{clk} \quad f_{Q_2} = \frac{1}{8} f_{clk}$$

$$f_{Q_3} = \frac{1}{16} f_{clk}$$

Advantages of Syn Counter over Asyn Counter

- The o/p of all the ff will change simultaneously.
- A syn counter can operate at much higher freq, but the circuitry are more complex than that of the syn counter.

Counter design steps

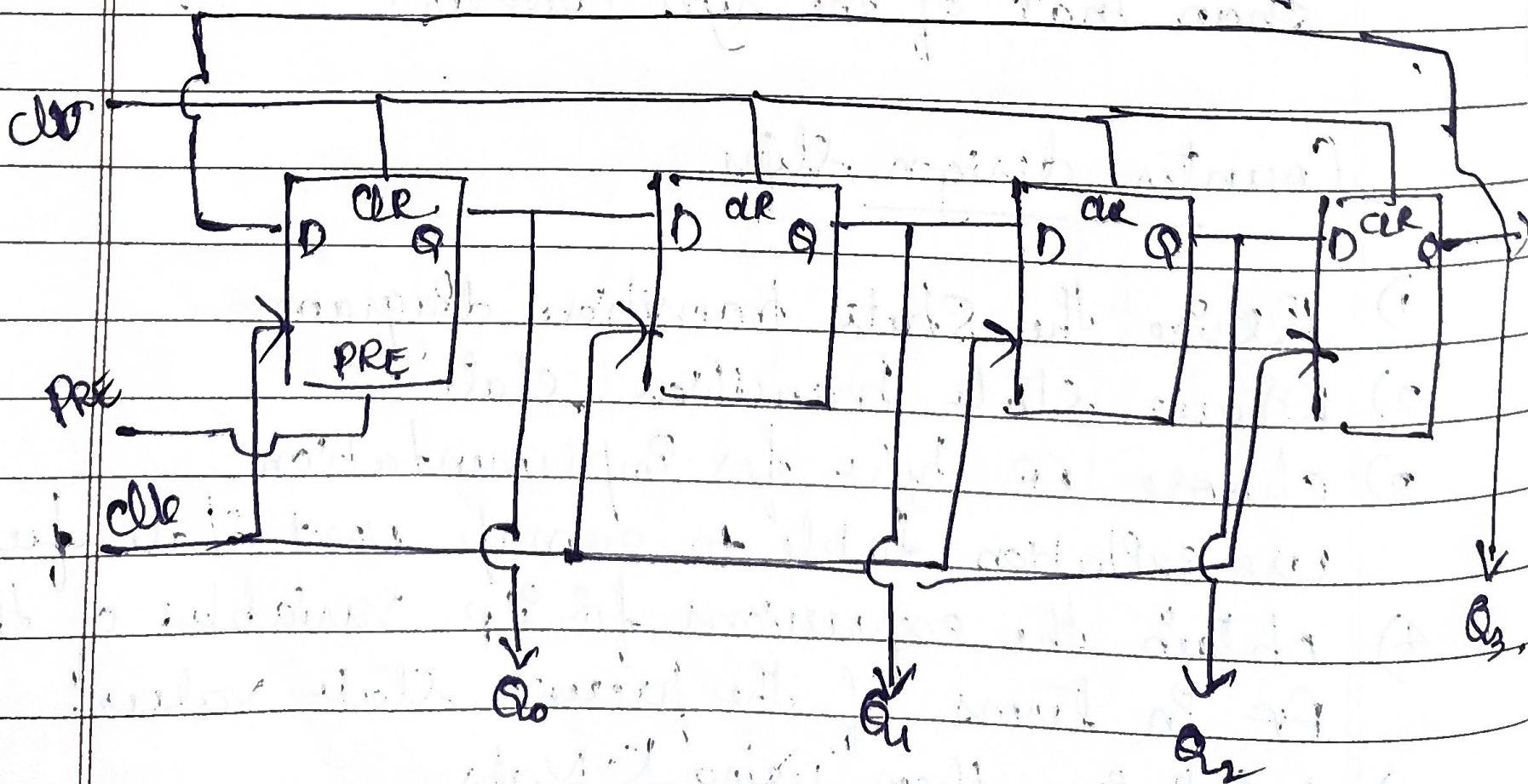
- 1) Draw the State transition diagram
- 2) Draw state transition table
- 3) Choose ff type for implementation.
Use excitation table to sumap next state functions
- 4) Obtain the expressions for i/p variables of the ff in terms of the present state values.
- 5) Minimize them using K-Maps

- 6) Implement Resulting Logic with minimized expressions.

Shift Register Counter

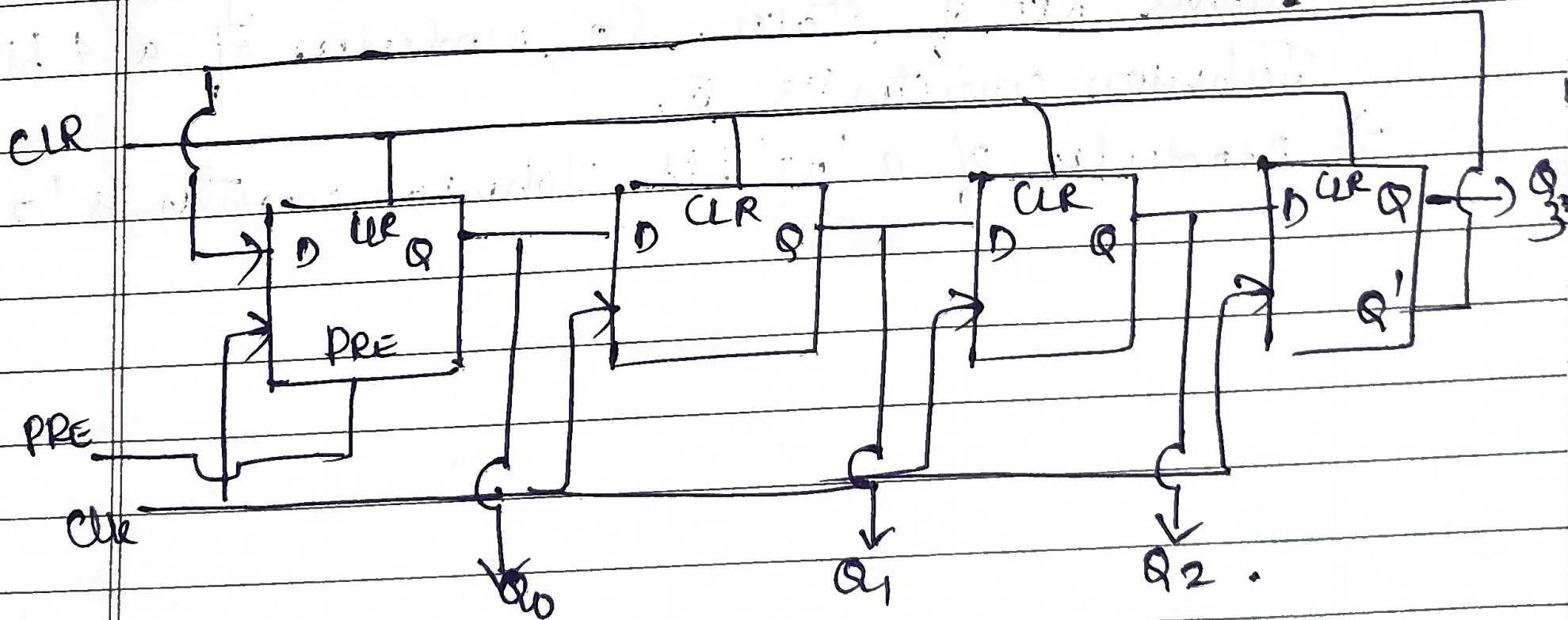
- Counters are used for producing a special sequence of numbers.
- Shift register with serial o/p connected back to the serial i/p to produce special sequence
- 2 types $\rightarrow$ Ring Counter
 $\rightarrow$ Johnson Counter

Ring Counter



Cycle no	Q ₀	Q ₁	Q ₂	Q ₃	
0	1	0	0	0	4 cycles
1	0	1	0	0	
2	0	0	1	0	
3	0	0	0	1	
4	1	0	0	0	4 cycles.
5	0	1	0	0	
6	0	0	1	0	
7	1	0	0	1	
8					

Twisted Ring Counter



Cycle no	Q_0	Q_1	Q_2	Q_3	Q_3'
0	1	0	0	0	1
1	1	1	0	0	1
2	1	1	1	0	1
3	1	1	1	1	0
4	0	1	1	1	0
5	0	0	1	1	0
6	0	0	0	1	0
7	0	0	0	0	1
8	1	0	0	0	1

→ for every 8 clock cycles, this counter goes into the same set of states. So, modulus of a 4 bit Johnson counter is 8.

→ Modulus of a 'n' bit Johnson counter is '2n'.